FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-053 ColdChain Lakehouse Mini-Lab

Fresh limited-hint integration after all 13 lesson mastery records.

Artifact	C3-053
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse | 13 lessons | 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6 | AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.

ColdChain scenario

ColdChain receives temperature-controlled shipment files. Day 2 corrects shipment SH002, inserts SH005 and adds a priority field. Build a small trusted lakehouse path that remains queryable, auditable, rerunnable and recoverable.

Architecture picture

source CSV evidence -> typed Parquet/layout -> trusted Delta table -> controlled schema change -> MERGE/reconcile -> prior-version read -> layer/catalog contracts -> maintenance/retention safety

Setup

Unzip C3_053_COLDCHAIN_MINI_LAB_STARTER_RC1.zip into a fresh workspace. Do not reuse your practice Delta table/output state.

Staged build

- State current-state grain: one trusted row per shipment_id.
- Choose/justify Parquet layout and partition approach.
- Create trusted Delta from day 1 and save real committed version evidence.
- Classify priority as a contract change before enabling supported additive evolution.
- Upsert SH002 + insert SH005; prove no duplicate shipment_id.
- Reconcile row count plus weight_kg and operation metrics.
- Read a prior committed version and name exact differences.
- Write bronze/silver/gold and logical table contracts.
- Define measurable compaction trigger plus retention/time-travel safety.
- Compare Delta and Iceberg conditionally using engine/catalog/DML/operations/interoperability factors.

Required competencies

- Physical layout / format / partition reasoning
- Real Parquet metadata and file-health evidence
- Transactional Delta version/log evidence
- Controlled schema evolution and contract change
- Grain-safe MERGE/upsert with reconciliation
- Historical version read and retention reasoning
- Layer/catalog/open-table-format architecture and maintenance handoff

Pass rule

Every required competency above must PASS and there must be zero unresolved critical failure. Missing real Parquet/Delta runtime proof cannot be averaged away.

Critical failures

- Duplicate current-state business keys remain.
- Parquet/Delta output is claimed without actual execution.
- Incompatible schema change is silently accepted.
- Required history is destructively removed just to finish the exercise.
- Layer/table contract cannot state grain/owner/replay/recovery.